

Direction-dependent strain transfer function of DAS considering gauge length and pulse shape with its implications on high spatial resolution acquisitions

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Summary

The use of quantitative distributed acoustic sensing (DAS) to measure both the amplitude and phase of dynamic ground strains has gained interest in geophysical and engineering communities. Applications like geo-structural health monitoring, seismic gradiometry, and full-waveform inversion (FWI) benefit from both amplitude and phase measurement information. DAS would be a convenient tool for these applications because of its large spatial coverage, trace density and exceptional strain sensitivity. However, the distributed nature of the measurements makes their usefulness limited, as the gauge length and optical pulse shape act as spatial lowpass filters. The combination of these filters determines the theoretical response of the DAS assuming the optical fiber is coupled with the host media. Here, the omnidirectional transfer functions accounting for gauge length and pulse shape are presented and their implications on high spatial resolution applications are discussed.

Introduction

Quantitative DAS measures the spatial average of dynamic strain experienced by an optical fiber. The approximate length over which the average is taken is called the gauge length, which can range from ones to tens of meters. The idealized gauge length assumes that the averaging window is perfectly rectangular, however the optical pulses that scatter from either end of the gauge length are not perfect impulses. This causes an averaging filter that is a smoothed rectangular window. As the gauge length decreases, the impact of the pulse shape on the overall instrument response increases. The total transfer function accounting for these effects can be described as a function of apparent wavenumber acting along the 1-D direction of a DAS fiber.

Apparent Wavenumber

To generalize the DAS transfer function, it is necessary to describe the wavenumber of any seismic wave as a projection onto a 1D measurement direction. Apparent wavenumber (k_a) can be defined as:

$$k_a = \mathbf{k} \cdot \hat{\mathbf{a}} \quad (1)$$

where $\mathbf{k}$ is the wave vector and $\hat{\mathbf{a}}$ is the unit vector in the direction of the DAS array.

DAS transfer function

For description of the DAS transfer function, the continuous strain field is considered as the input and DAS-measured strain field is the output. The DAS transfer function relative to apparent wavenumber ($H(k_a)$) can then be represented assuming perfect coupling with the host media as:

$$H(k_a) = \mathbb{G}(k_a) \mathbb{P}(k_a) \quad (2)$$

where $\mathbb{G}(k_a)$ and $\mathbb{P}(k_a)$ are the transfer functions due to gauge length and pulse shape, respectively. The gauge length transfer function can be described as the Fourier transform of a scaled rectangular window as:

$$\mathbb{G}(k_a) = \mathcal{F} \left\{ \frac{1}{g} \Pi\left(\frac{x}{g}\right) \right\} = \frac{\sin(\pi k_a g)}{\pi k_a g} \quad (3)$$

where g is the gauge length of the DAS system, $\Pi(x)$ is the rectangular window (Bracewell, 1986) and x is the ordinate along the $\hat{\mathbf{a}}$ direction. This has been described previously for the case of axially incident waves by Dean, 2017. The pulse shape can vary dependent on the optoelectronics of a DAS system. One possibility is that the pulses are approximately shaped as Gaussian windows (Dean, 2016). This can be described as:

$$\mathbb{P}(k_a) = \mathcal{F} \left\{ \frac{1}{\sqrt{2\pi}\sigma_x} e^{-\frac{x^2}{2\sigma_x^2}} \right\} = e^{-\frac{k_a^2 \sigma_x^2}{2}} \quad (4)$$

σ_x is related to the pulse full width at half-maximum (h_p) by:

$$\sigma_x = \frac{h_p}{2.3548} \quad (5)$$

and σ_{k_a} is related to σ_x by:

$$\sigma_k = \frac{1}{2\pi\sigma_x} \quad (6)$$

However, it is also possible to make approximately rectangular optical pulses (Wang et al. 2015) or even sinc-shaped pulses (Soto et al 2013). For these reasons, (2) can be left general with respect to pulse shape:

$$H(k_a) = \frac{\sin(\pi k_a g)}{\pi k_a g} \mathbb{P}(k_a) \quad (7)$$

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Implications for field acquisitions

Since $H(k_a)$ is a multiplication of two transfer functions, the total response may be dominated by one or the other. DAS users may have the capability to change the gauge length of their acquisitions, but pulse shape is typically fixed by manufacturers. The gauge length and pulse shape effects are shown in Figure 1 for a gauge length of 2 m and Gaussian pulse with $h_p = 2$ m. The trace separation determines the Nyquist frequency of the transfer function. In the following cases it is taken as 1 m ($k_{ny} = 0.5 \frac{1}{m}$).

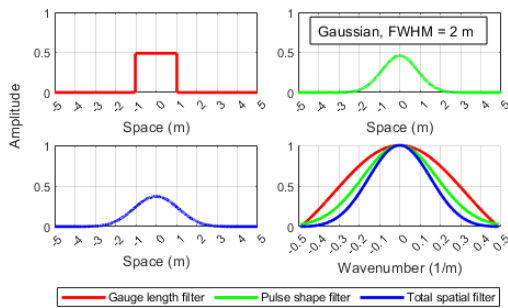


Figure 1. Filters induced by the gauge length, pulse shape and combination of gauge length and pulse shape for a gauge length of 2 m and an $h_p = 2$ m-Gaussian pulse.

If the gauge length is reduced in the limiting sense, it becomes an impulse. This case is shown in Figure 2.

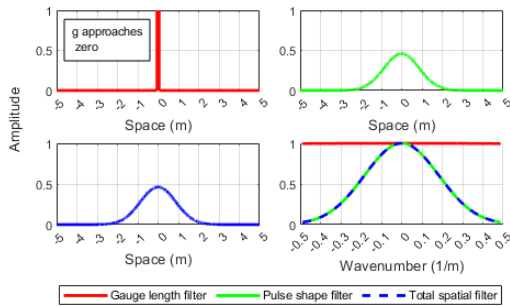


Figure 2. Filters induced by the gauge length, pulse shape and combination of gauge length and pulse shape for a gauge length that approaches zero and a $h_p = 2$ m-Gaussian pulse.

Reducing the gauge length has negligible benefit in this case because the pulse shape will limit the transfer function. The case of a rectangular pulse of 1 m width implemented with a 1 m gauge length is shown in Figure 3.

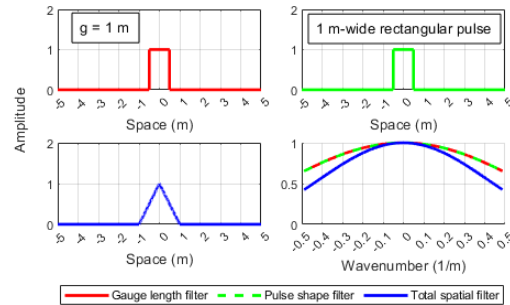


Figure 3. Filters induced by the gauge length, pulse shape and combination of gauge length and pulse shape for a gauge length of 1 m and a rectangular pulse 1 m wide.

In this case the gauge length and pulse shape would impart the same effect as each other on the measured data.

Conclusions

The idealized DAS transfer function can be formulated using the apparent wavenumber. When using DAS for high spatial resolution applications, the pulse shape and gauge length are equally important for determining the magnitude response of the system. It is encouraged that practitioners of DAS and manufacturers communicate and consider the pulse shape equally to the gauge length so that the magnitude response can be quantified. Field studies should be done to evaluate the instrument response due to different pulse shapes and validate these theoretical findings experimentally.

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